Lecture 2. Axioms of Probability

Sample spaces, axioms, inclusion–exclusion, equally likely outcomes, continuity of probability

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2.1 Sample space and events

The **sample space** S is the set of outcomes; an **event** is a subset. We use $E \cup F$, $EF = E \cap F$, $E^c = S \setminus E$, $\emptyset$; E, F are **mutually exclusive** if $EF = \emptyset$. Union and intersection are commutative, associative and mutually distributive, and

$$\Big(\bigcup_{i=1}^{n} E_i\Big)^c = \bigcap_{i=1}^{n} E_i^c, \qquad \Big(\bigcap_{i=1}^{n} E_i\Big)^c = \bigcup_{i=1}^{n} E_i^c \qquad \text{(De Morgan)}.$$

(An outcome lies outside every E_i iff it lies outside their union; apply this to the E_i^c for the second.)

2.2 Axioms of probability

Relative frequency $\lim n(E)/n$ motivates but does not define probability; we impose axioms.

Definition (Probability)

A function P on events is a **probability** if (i) $0 \le P(E) \le 1$;
(ii) $P(S) = 1$;
(iii) for mutually exclusive $E_1, E_2, \ldots$, $P\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} P(E_i)$.

$E_1 = S$, $E_i = \emptyset$ $(i \ge 2)$ forces $P(\emptyset) = 0$, and (iii) then gives finite additivity. (For uncountable S, P is defined on a suitable class of events.)

Proposition 2.1

(a) $P(E^c) = 1 - P(E)$.
(b) $E \subset F \Rightarrow P(E) \le P(F)$.
(c) $P(E \cup F) = P(E) + P(F) - P(EF)$.

Proof. (a) E and E^c are disjoint with union S.

(b) $F = E \cup E^c F$, a disjoint union.

(c) $E \cup F = E \cup E^c F$ and $F = EF \cup E^c F$, both disjoint; subtract. ∎

Theorem 2.2 (Inclusion–exclusion)

$$P\Big(\bigcup_{i=1}^{n} E_i\Big) = \sum_i P(E_i) - \sum_{i<j} P(E_i E_j) + \ldots + (-1)^{r+1} \sum_{i_1<\cdots<i_r} P(E_{i_1} \cdots E_{i_r}) + \ldots + (-1)^{n+1} P(E_1 \cdots E_n).$$

Proof. For countable S, check each outcome contributes equally to both sides: one in no E_i contributes 0; one in exactly $m \ge 1$ contributes once on the left and $\binom{m}{1} - \binom{m}{2} + \cdots \pm \binom{m}{m} = 1 - \sum_{i=0}^{m} (-1)^i \binom{m}{i} = 1$ times on the right. (General S: induct with Proposition 2.1(c).) ∎

Truncations alternate between upper and lower bounds (Bonferroni):

$$P\Big(\bigcup E_i\Big) \le \sum P(E_i), \qquad P\Big(\bigcup E_i\Big) \ge \sum P(E_i) - \sum_{i<j} P(E_i E_j), \quad \ldots$$

2.3 Equally likely outcomes

If $S = \{1, \ldots, N\}$ with equal probabilities, $P(E) = |E|/N$. Treating objects as distinguishable never changes the answer and makes outcomes equally likely.

Example 2.1

(a) Three balls from 6 white and 5 black: $P(\text{one white, two black}) = \binom{6}{1}\binom{5}{2}/\binom{11}{3} = 4/11$; ordered draws give $3 \cdot 6 \cdot 5 \cdot 4/(11 \cdot 10 \cdot 9) = 4/11$.
(b) k of n balls drawn, one special: by symmetry $P(\text{special drawn}) = k/n$; directly $\binom{n-1}{k-1}/\binom{n}{k} = k/n$.
(c) Poker full house (rank and suits of the triple, then of the pair): $13\binom{4}{3} \cdot 12\binom{4}{2}/\binom{52}{5} = 3744/2{,}598{,}960 \approx 0.00144$.

Example 2.2 (Birthday problem)

With all 365^n assignments equally likely, n people have distinct birthdays with probability $365 \cdot 364 \cdots (366 - n)/365^n$; a shared birthday has probability 0.507 at $n = 23$ and 0.970 at $n = 50$. (23 people form 253 pairs, each matching with probability $1/365$.)

Example 2.3 (The card after the first ace)

A shuffled deck is turned up. Is the card after the first ace more likely the ace of spades or the two of clubs? Remove the ace of spades, order the other 51 cards (51! ways) and reinsert it right after the first ace: probability $51!/52! = 1/52$. The same holds for the two of clubs.

Example 2.4 (The matching problem)

N people pick hats at random; E_i: person i gets their own. $P(E_{i_1} \cdots E_{i_r}) = (N - r)!/N!$ for each of $\binom{N}{r}$ index sets, so

$$P\Big(\bigcup_i E_i\Big) = \sum_{r=1}^{N} (-1)^{r+1} \binom{N}{r} \frac{(N-r)!}{N!} = \sum_{r=1}^{N} \frac{(-1)^{r+1}}{r!},$$

and

$$P(\text{no match}) = \sum_{r=0}^{N} \frac{(-1)^r}{r!} \xrightarrow[N\to\infty]{} e^{-1} \approx 0.368.$$

Exactly k matches (choose them, no match among the other $N - k$):

$$P(\text{exactly } k) = \binom{N}{k} \frac{(N-k)!}{N!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} = \frac{1}{k!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} \approx \frac{e^{-1}}{k!},$$

a Poisson(1) probability (Lecture 4).

Example 2.5 (Couples at a round table)

Ten couples at a round table (19! circular orders); E_i: couple i together. Gluing r couples leaves $20-r$ units in $(19-r)!$ circular orders, each block in 2 ways:

$$P(\text{no couple together}) = \sum_{r=0}^{10} (-1)^r \binom{10}{r} \frac{2^r (19-r)!}{19!} \approx 0.3395.$$

2.4 Probability as a continuous set function

Motivation. Countable additivity makes probability respect monotone limits such as "eventually".

A sequence of events is **increasing** if $E_n \subset E_{n+1}$, with limit $\bigcup E_n$, and **decreasing** if $E_n \supset E_{n+1}$, with limit $\bigcap E_n$.

Proposition 2.3 (Continuity)

For an increasing or decreasing sequence, $\lim_n P(E_n) = P(\lim_n E_n)$.

Proof. Increasing: $F_1 = E_1$, $F_n = E_n E_{n-1}^c$ are disjoint with $\bigcup_1^n F_i = E_n$, $\bigcup_1^\infty F_i = \bigcup_1^\infty E_i$, so by (iii)

$$P\Big(\bigcup_i E_i\Big) = \sum_{i=1}^{\infty} P(F_i) = \lim_n \sum_{i=1}^{n} P(F_i) = \lim_n P(E_n).$$

Decreasing: apply this to E_n^c and use De Morgan. ∎

Example 2.6

A fair coin is tossed forever. E_n (first n tosses tails) decreases to T (no head ever), so $P(T) = \lim 2^{-n} = 0$: a head appears with probability 1, though "all tails" is possible.

Coherent subjective probabilities obey the same axioms.

2.5 Selected exercises

Motivation. Ross's theoretical exercises (book numbering): inclusion–exclusion by induction, Bonferroni, and two counting facts.

Exercise 2.8 (Counting partitions)

Let T_n be the number of partitions of $\{1, \dots, n\}$ into nonempty blocks, and put $T_0 = 1$.
(a) Show that $T_3 = 5$.
(b) Show that $T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k$ for $n \geq 1$, and deduce $T_4 = 15$.

Solution.

(a) The partitions of $\{1,2,3\}$ are $\{123\}$, $\{1\}\{23\}$, $\{2\}\{13\}$, $\{3\}\{12\}$ and $\{1\}\{2\}\{3\}$.

(b)

Step 1. A partition π of $\{1, \dots, n+1\}$ determines the block $B \ni n+1$ and the partition π' of $R = \{1, \dots, n\} \setminus B$ by the other blocks (empty if $R = \emptyset$); conversely $A \subseteq \{1, \dots, n\}$ and a partition π' of the complement give $\{A \cup \{n+1\}\} \cup \pi'$. These are inverse maps.

Step 2. With $k = |R|$: $\binom{n}{k}$ choices of A, then T_k choices of π' ($k \geq 1$) or one ($k = 0$):

$$T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k.$$

Step 3. $T_4 = 1 + 3T_1 + 3T_2 + T_3 = 1 + 3 + 6 + 5 = 15$ (then $T_5 = 52, \ldots, T_{10} = 115{,}975$). ■

Exercise 2.14 (Inclusion–exclusion by induction)

Prove Theorem 2.2 for an arbitrary sample space by induction on n.

Solution. With $E_I = \bigcap_{i \in I} E_i$, the claim is $P(\bigcup_{i=1}^n E_i) = \sum_I (-1)^{|I|+1} P(E_I)$ over nonempty $I \subseteq \{1, \ldots, n\}$.

Step 1. $n = 1$ is trivial; $n = 2$ is Proposition 2.1(c).

Step 2. Assume the claim for n events; let $U = \bigcup_{i=1}^n E_i$. By Proposition 2.1(c) and $UE_{n+1} = \bigcup_{i=1}^n (E_i E_{n+1})$,

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = P(U) + P(E_{n+1}) - P\Big(\bigcup_{i=1}^{n} E_i E_{n+1}\Big).$$

Step 3. Apply the hypothesis to $E_1, \ldots, E_n$ and to $E_1 E_{n+1}, \ldots, E_n E_{n+1}$, whose intersection over J is $E_{J \cup \{n+1\}}$:

$$\begin{aligned} P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = &\sum_J (-1)^{|J|+1} P(E_J) + P(E_{\{n+1\}}) \\ &- \sum_J (-1)^{|J|+1} P(E_{J \cup \{n+1\}}), \end{aligned}$$

both sums over nonempty $J \subseteq \{1, \ldots, n\}$.

Step 4. Each nonempty $I \subseteq \{1, \ldots, n+1\}$ occurs once: $n+1 \notin I$ in the first sum, sign $(-1)^{|I|+1}$; $I = \{n+1\}$ as the middle term, sign $+1$; $I = J \cup \{n+1\}$ in the last sum, sign $-(-1)^{|J|+1} = (-1)^{|I|+1}$. ■

Exercise 2.16 (Bonferroni's inequality)

Show that for any events $E_1, \ldots, E_n$,

$$P(E_1 E_2 \cdots E_n) \geq P(E_1) + \cdots + P(E_n) - (n-1).$$

Solution.

Step 1. ($n = 2$.) $P(EF) = P(E) + P(F) - P(E \cup F) \geq P(E) + P(F) - 1$. ($n = 1$ is an equality.)

Step 2. (Induction.) Apply Step 1 to $E = E_1 \cdots E_n$, $F = E_{n+1}$:

$$\begin{aligned} P(E_1 \cdots E_{n+1}) &\geq P(E_1 \cdots E_n) + P(E_{n+1}) - 1 \\ &\geq \sum_{i=1}^{n} P(E_i) - (n-1) + P(E_{n+1}) - 1 = \sum_{i=1}^{n+1} P(E_i) - n. \end{aligned}$$

E.g. $P(E) = 0.9$, $P(F) = 0.8$ give $P(EF) \geq 0.7$ under any dependence. ■

Exercise 2.19 (Waiting for the r-th red ball)

An urn contains n red and m blue balls. They are withdrawn one at a time, at random, until a total of $r \le n$ red balls have been withdrawn. Find the probability that exactly k balls are withdrawn.

Solution.

Step 1. Take the $(n+m)!$ equally likely orders of all (labelled) balls. Exactly k are withdrawn iff the first $k-1$ positions hold exactly $r-1$ red and position k is red ($r \le k \le r+m$).

Step 2. Count: the balls in the first $k-1$ positions ($\binom{n}{r-1}\binom{m}{k-r}$) and their order ($(k-1)!$), the red ball at k ($n-r+1$), and the order of the rest ($(n+m-k)!$).

Step 3. Divide by $(n+m)!$, using $\binom{n+m}{k-1}^{-1} = (k-1)!\,(n+m-k+1)!/(n+m)!$:

$$P(k \text{ balls withdrawn}) = \frac{\binom{n}{r-1}\binom{m}{k-r}}{\binom{n+m}{k-1}} \cdot \frac{n-r+1}{n+m-k+1}, \qquad r \le k \le r+m.$$

(hypergeometric probability of $r-1$ red in $k-1$ draws, times that the next is red). ■

Exercise 2.20 (No uniform distribution on a countable set)

Let the sample space $S = \{s_1, s_2, \dots\}$ be countably infinite. Show that not all outcomes can be equally likely. Can all outcomes have positive probability?

Solution.

Step 1. By (ii) and (iii), $1 = P(S) = \sum_{i=1}^{\infty} P(\{s_i\})$.

Step 2. If all $P(\{s_i\}) = p$, the series is 0 ($p = 0$) or ∞ ($p > 0$), never 1.

Step 3. Yes: $P(E) = \sum_{i:s_i \in E} 2^{-i}$ is nonnegative, has $P(S) = 1$, and is countably additive (nonnegative series can be regrouped). ■